

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 14

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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- 1** A polynomial of degree 5 with real coefficients has how many real roots at least, and at most?
- A** at least 1, at most 5
 - B** exactly 5
 - C** at least 1, at most 4
 - D** at least 0, at most 4
 - E** at least 0, at most 5
 - F** at least 2, at most 5

2 The value of a car falls by 20% each year. After how many complete years is it first worth less than half its original value?

A 5

B 6

C 4

D 10

E 2

F 3

3 Simplify $\sqrt{48} \times \sqrt{27} \div \sqrt{12}$.

A 18

B 12

C $9\sqrt{3}$

D 6

E 36

F $6\sqrt{3}$

- 4 TA is a tangent to a circle at A , and AB is a chord. C is a point on the circle on the opposite side of AB from T , with angle $ACB = 65^\circ$. Find angle TAB .

- A 25°
- B 65°
- C 115°
- D 130°
- E 50°
- F 32.5°

- 5 The displacement s metres of a particle at time t seconds is $s = t^3 - 6t^2 + 9t$. At what times is the particle momentarily at rest?
- A** $t = 2$ only
- B** $t = 1$ and $t = 3$
- C** $t = 1$ only
- D** $t = 3$ only
- E** $t = 0$ and $t = 3$
- F** $t = 0, t = 1$ and $t = 3$

6 Which of the following is equal to $\frac{8^4 \times 2^{-5}}{4^3}$?

A 1

B 2

C 16

D 8

E 4

F $\frac{1}{2}$

7 Find the quotient when $2x^3 - 3x^2 + 4x - 5$ is divided by $(x - 1)$.

A $2x^2 + x - 3$

B $2x^2 - 5x + 9$

C $2x^2 - x + 3$

D $2x^2 - x - 3$

E $2x^2 + x + 3$

F $2x^2 - x + 5$

8 When $x^3 - 4x^2 + kx + 6$ is divided by $(x - 3)$ the remainder is -6 . Find k .

A 1

B 5

C -3

D -5

E 3

F -1

9 The quadratic $y = ax^2 + bx + c$ has its vertex at $(2, -3)$ and passes through $(0, 5)$. Find $a + b + c$.

A -1

B 7

C 1

D 5

E -3

F 3

10 The points $(1, 7)$ and $(5, -1)$ are the ends of a diameter of a circle. Find the equation of the circle.

A $(x - 3)^2 + (y - 3)^2 = 40$

B $(x - 2)^2 + (y - 4)^2 = 20$

C $(x - 3)^2 + (y - 3)^2 = \sqrt{20}$

D $(x - 3)^2 + (y - 3)^2 = 20$

E $(x - 3)^2 + (y - 3)^2 = 80$

F $(x + 3)^2 + (y + 3)^2 = 20$

- 11** Find $\int \left(x + \frac{1}{x}\right)^2 dx$.
- A** $\frac{x^3}{3} + 2x - \frac{1}{x} + c$
- B** $\frac{3}{x^3} - \frac{1}{x} + c$
- C** $\frac{1}{3^3} \left(x + \frac{1}{x}\right)^3 + c$
- D** $\frac{x^3}{3} + 2x + \frac{1}{x} + c$
- E** $\frac{3}{x^3} + 2x + c$
- F** $x^3 + 2x - \frac{1}{x} + c$

- 12** The function f is defined by $f(x) = |x - 2| + |x + 2|$ for all real x . What is the minimum value of $f(x)$, and for how many values of x is it attained?
- A** minimum 0, attained for exactly two x
 - B** minimum 2, attained for exactly one x
 - C** minimum 4, attained for exactly two x
 - D** minimum 4, attained for infinitely many x
 - E** minimum 4, attained for exactly one x

- 13** A quantity halves every 5 hours. Which expression gives the number of hours for it to fall to $\frac{1}{10}$ of its initial value?
- A** $5\log_2 5$
- B** $\frac{\log_{10} 2}{5}$
- C** $5\log_{10} 2$
- D** $\frac{5}{\log_{10} 2}$
- E** 25
- F** $50\log_{10} 2$

- 14** The graph of $y = \sin x$ (with x in radians) crosses the line $y = x$ at how many points?
- A** 2
 - B** 3
 - C** 0
 - D** 1
 - E** infinitely many

- 15** The solution set of $ax^2 + bx + c < 0$ (with $a > 0$) is $-2 < x < 5$. What is the solution set of $ax^2 - bx + c < 0$?
- A** $-2 < x < 5$
- B** $x < -5$ or $x > 2$
- C** $2 < x < 5$
- D** $x < -2$ or $x > 5$
- E** $-5 < x < -2$
- F** $-5 < x < 2$

16 The roots of $x^2 - 5x + 2 = 0$ are α and β . Find the value of $\alpha^2 + \beta^2$.

A 25

B 29

C 23

D 27

E 21

F 17

17 A square-based pyramid has base side 6 and all slant edges of length 9. Find its height.

A $\sqrt{81 - 36}$

B $\sqrt{45}$

C 6

D $\sqrt{63}$

E $\sqrt{72}$

F $3\sqrt{6}$

- 18** The function f satisfies $f(x) = x^2$ for all x . Which of the following transformations maps $y = f(x)$ onto $y = 4x^2 - 8x + 4$?
- A** a translation of 1 unit in the positive x -direction followed by a stretch parallel to the y -axis with factor 4
 - B** a translation of 2 units in the positive x -direction followed by a stretch parallel to the y -axis with factor 2
 - C** a translation of 1 unit in the negative x -direction followed by a stretch parallel to the y -axis with factor 4
 - D** a stretch parallel to the y -axis with factor 4 followed by a translation of 4 units in the positive y -direction
 - E** a stretch parallel to the x -axis with factor 2 followed by a translation of 2 units in the positive x -direction

- 19** Two circles of radius 2 have their centres a distance 2 apart. Find the area of the region common to both circles.

- A** $\frac{2\pi}{3}$
B $\frac{4\pi}{3} - 2\sqrt{3}$
C $\frac{8\pi}{3} - 2\sqrt{3}$
D $\frac{8\pi}{3} - 4\sqrt{3}$
E $2\pi - 2\sqrt{3}$
F $\frac{4\pi}{3} - \sqrt{3}$

20 How many solutions does the equation $\sin x = \frac{x}{10}$ have?

A 7

B 6

C 1

D 5

E 9

F 3

END OF TEST